

Course Code : 2104CS352

Date : 24-05-2024

Course Name : Internet of Things

Duration : 150 Minutes

Total Marks : 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

- Q.1 (A)** Recite Enabling technologies in IoT. **4**
- (B)** Define actuators with example. **3**

OR

Define IoT and its characteristics.

- (C)** Discuss IoT challenges in detail. **7**

OR

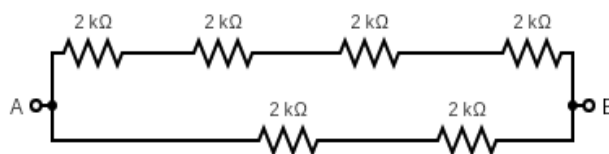
Explain 7-layer IoT architecture with diagram.

- Q.2 (A)** Define Resistor, Capacitor and Inductor in electronic circuit. **4**
- (B)** Isolate open circuit, close circuit and Short circuit. **3**

OR

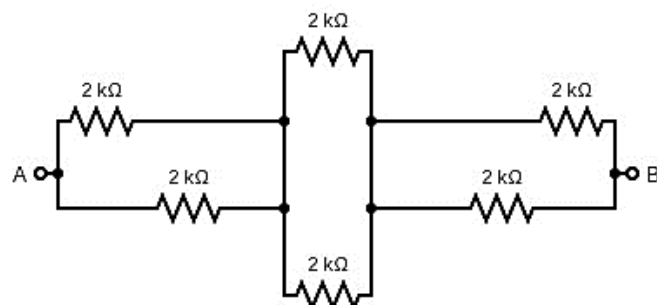
Define Ohm's law for the Electrical circuit.

- (C)** Find the equivalent resistance- R_{AB} between node A and B. **7**



OR

Find the equivalent resistance- R_{AB} between node A and B



- Q.3 (A)** Define Relay with its five pin. **4**
- (B)** Write an Arduino code to toggle LED with 5 second delay. **3**

OR

Write an Arduino code to continuous print "Hello Arduino" on serial monitor, with 2 second delay.

- (C)** Write an Arduino code to measure Light intensity inside the room, with Arduino UNO interfacing circuit. **7**

OR

Write an Arduino code to measure distance using Ultrasonic sensor. (Speed of Ultrasonic Sound = 0.0345 cm/micro-sec)

- Q.4 (A)** Recall map(),max(),min() and pow() functions in Arduino. **4**
- (B)** Discuss analog port in Arduino UNO. **3**

OR

Define Pulse Width Modulation and explain analog out in Arduino.

- (C)** Demonstrate read and Write Arduino functions for analog and digital signals. **7**

OR

Sketch the Arduino UNO board architecture in detail.

- Q.5 (A)** Define common architecture of IoT application for the agriculture sector. **4**
- (B)** Discuss BLYNK_WRITE() function to read data from Blynk cloud dashboard. **3**

OR

Discuss Blynk.virtualWrite() function to Update data on the dashboard (Blynk IoT cloud).

- (C)** Explain Fog computing and compare it with cloud computing. **7**

OR

Explain Iaas, Paas and Saas with respect to Cloud Computing.
